A-A-50214
3 August 1983

COMMERCIAL ITEM DESCRIPTION

BALLAST REGULATOR

RAILWAY, MAINTENANCE-OF-WAY, 56-1/2 INCH GAGE

This proposed Commercial Item Description draft, as prepared by the Mobility Equipment Research and Development Command, Fort Belvoir, VA 22060, has not been approved for promulgation and is subject to modification. DO NOT USE FOR ACQUISITION PURPOSES.

This Commercial Item Description covers the requirements for a ballast regulator meeting all applicable American Railway Engineering Association (AREA) requirements.

Salient characteristics:

Minimum weight of unit with all attachments should not be less than 30,000 lbs.
Maximum height of unit shall be 10'6".
Standard commercial diesel engine sized to meet performance requirements set forth under "intended use". Engine shall be enclosed with movable panels for ease of maintenance.
Transmission capable of transmitting the full power of the engine with equal speed in either direction of travel and with all gear lever positions clearly and durably labeled.
Electric starting.
Fuel tank capacity shall be sized to allow 16 hours of continuous operation.
Industrial strength battery.
Heavy-duty alternator.
Fail-safe brakes applied at all four wheels.
Emergency brake system.
Elastomer or spring-type suspension.
Heavy duty tapered roller bearings.
Flanged steel wheels meeting AAR requirements.
Enclosed pressurized cab fully insulated for heat/cold and sound.
Cab heater and air conditioner.
All windows shall be distortion-free safety glass and provide 360 degree visibility.
Upholstered seating for at least two persons.
Interior cab light.
Sealed beam headlights and taillights front and rear.
Capability for night operation.
Strobe beacon light on roof.
Illuminated instrument panel equipped with standard operating controls, gages, and switches and with all positions and functions clearly labeled.
Electrohydraulic controls.
Electric windshield wiper front and rear.
Horn.

FSC 2230

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Storage space for tools.
One 5-pound dry chemical fire extinguisher secured in the cab with location clearly stencilled.
Front or center directional split plow.
Adjustable side wings capable of reclaiming ballast to 10'0" minimum from centerline of track.
Two ballast boxes of 3/4 c.y. minimum capacity each.
Scarifying teeth, rooter tooth, and tie end plow.
Ballast broom with stone shield capable of sweeping at a rate of at least 2 miles per hour. Sweeping elements of rubber-covered steel cable.
Turntable.
Four lifting lugs of sufficient capacity to lift ballast regulator with location clearly indicated by stencilling "SLING HERE".
Semi-gloss enamel paint system in AREA standard yellow with black stencilling (except color-coding of tanks per AREA requirements).
Permanent metal identification plate mounted in a conspicuous place and showing Manufacturer's Name, Address, Model, Serial Number, and Date of Manufacture.

Intended use. The ballast regulator shall be capable of arranging ballast during maintenance-of-way operations and shall be designed for one-man operation with good visibility of all plowing operations and all functions controlled from within the cab. The plow and wings shall be adjustable and reversible for plowing in, plowing out, and transferring ballast from one shoulder to the other by a method not to exceed one pass in each direction of travel and while meeting various track profiles. The machine shall also be capable of deweeding, scarifying cemented ballast, and plowing material at tie ends to a minimum depth of 12 inches below bottom of ties. The ballast broom shall sweep tie tops, fill crib areas, and remove excess ballast to shoulders. The ballast regulator is intended to travel to and from work sites at a maximum self-propelled travel speed of not less than 30 mph on straight and level track. The ballast regulator shall be capable of negotiating a vertical grade of 3 percent and a horizontal curve of 100 feet radius.

Materials and workmanship. Materials shall meet ASTM and AREA specifications where applicable for the purpose intended. The ballast regulator shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed. Dissimilar metals, as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Workmanship in construction and finish shall be of high quality and consistent with industry standards.

Environmental condition. The ballast regulator shall be capable of operation without malfunction or damage at ambient temperatures from -25° F to +125 °F and under dusty atmospheric conditions.

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Essential safety features. All parts that rotate, reciprocate, or are subject to high operational temperatures and that are so located as to be a hazard to operating personnel, shall be guarded or insulated. All floors and steps shall have anti-skid surfaces. Fluid level indicators for engine, transmission, and other components shall be within easy and safe access of the operator or maintenance personnel. All controls shall be within convenient reach and vision of operator in his normal operating position and in accordance with SAE Human Factors Engineering Recommendations. Noise levels shall meet OSHA/FRA standards.

Standard commercial product. The above salient characteristics are set forth as minimum requirements. The ballast regulator shall be in accordance with these minimum requirements and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the model being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model. It shall have been in production, marketed, and in use for a minimum of one year. The introduction of normal product improvements is acceptable.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product sold in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Acceptance. Acceptance of the ballast regulator shall be based on visual inspection and demonstration of its performance capabilities. It shall be completely inspected to insure that all components have been provided and shall be given a 1 hour minimum continuous track run test combining forward and reverse directions at the highest speed practicable while operating all functional elements.

Packaging and marking. The items shall be preserved in accordance with the manufacturer's standard practice in a manner to prevent deterioration and damage. The ballast regulator shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

All doors and latches shall be secured. All windows shall be protected on the exterior side with exterior-type plywood or hardboard allowing a minimum one-inch clearance between the glass and the barrier. All systems shall be shipped with all fluid levels in normal operating ranges (except fuel). Antifreeze, if applicable, shall be added to provide protection for stated environmental conditions. A metal, wood, or composition plate of sufficient size to contain the marking information shall be attached in a conspicuous location. The

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marking shall consist of the item nomenclature, stock number, contract number, consignee's complete address and additional markings as specified in the contract or order. Markings shall be applied in a legible manner, in contrasting color, of an appropriate size, and a waterproof material.

Custodians
Army - ME
Navy - YD
Air Force - 84

PREPARING ACTIVITY:
Army - ME

Project 2230-0067

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